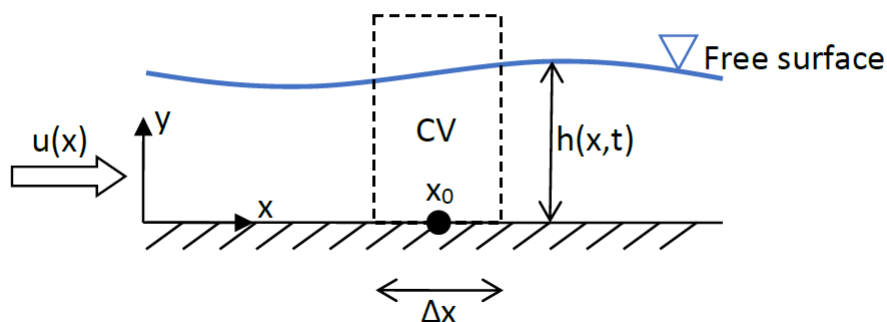


Practice Problems: Set - 1

Problem 1

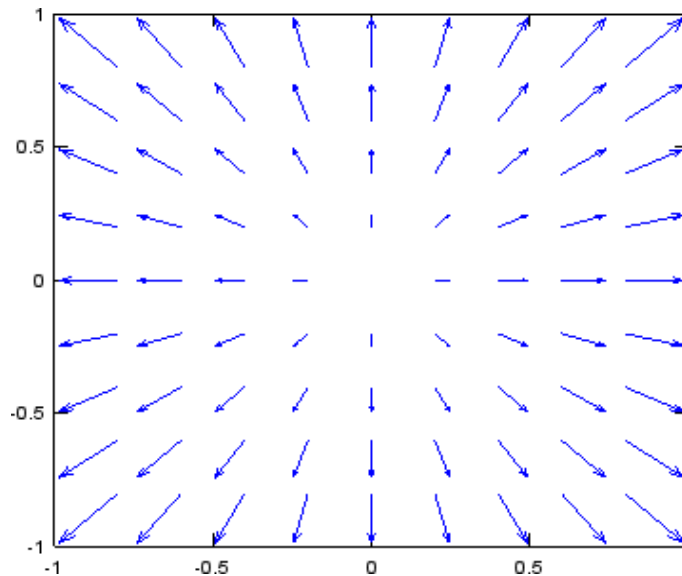
Consider the free surface flow shown in the figure below. The flow is from left to right and is bounded by the solid surface below and the free surface above. The free surface is exposed to the surrounding passive atmosphere (think of a river flow for example) and is described by $h(x,t)$, a function of location x and time t .

1. Remembering that the free surface can be thought of as a curve along which $f(x,y)=y-h=0$, and using the definition of the gradient, find an expression for the unit normal vector to the free surface.
2. In your own words, describe the statement of mass conservation applied to the control volume CV shown in the figure below.
3. Assuming a unidirectional and uniform velocity profile throughout derive the continuity equation for this problem if the fluid is incompressible. In other words, state the conservation of mass for CV in terms of h and u .
4. How would this mass conservation statement differ if the solid underneath the fluid was porous such that some of the fluid percolates through the solid surface at a rate of Q kg/m²/s. Check that your resulting equation is dimensionally correct.



Problem 2

A flow is planar, steady, incompressible, and purely radial as illustrated in the figure below. In other words, the only non-zero component of the velocity



vector in a polar coordinate system is u_r .

- Drawing a suitable control volume and clearly indicating the fluxes, derive an equation which represents the conservation of mass. This equation should be in terms of the radial velocity u_r and the distance from the axis r .
- Can the vector field above represent an incompressible flow?

Problem 3

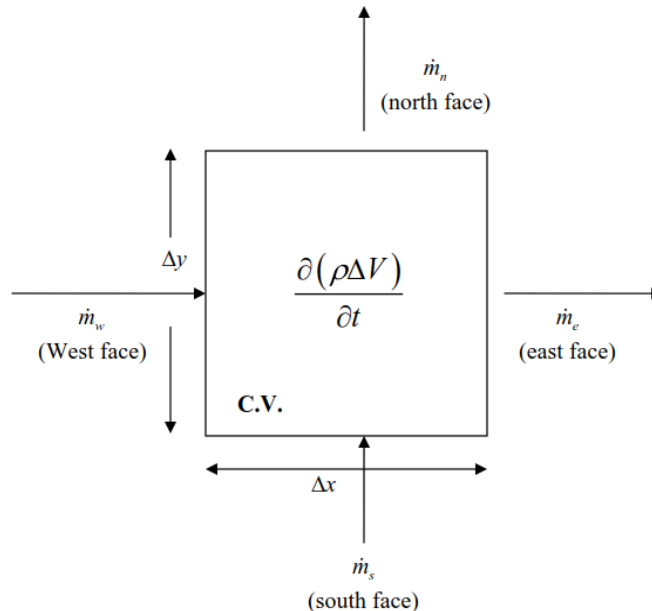
- What is the meaning of the substantial derivative?
- Write down the substantial derivative of the temperature of the temperature T .
- A small solid particle travels with a velocity of 2 m/s in the x-direction through a steady temperature field which increases linearly at a rate of 0.1 K/m in the x-direction. What will be the rate of change of the particle temperature?

Problem 4

1/ A fluid particle travels with constant velocity $(u, v, w) = (5, 5, 0)$ m/s in a room where the temperature is given by $T(x, y, z, t) = 10xe^{-2t}$.

- Find an expression for the rate of change of the temperature experienced by the fluid particle.
- Calculate the rate of change of the temperature of the fluid particle at $(x, y, z) = (1, 0, 0)$ and $t = 1$.

2/ Derive the continuity equation for any fluid accounting for the balance/budget of mass flowing in and out of the control volume (C.V.) shown below:



where ΔV is the volume of the control volume and $\dot{m}$ is the mass flow rate. Write the resulting equation in terms of the substantial derivative $D\rho/Dt$.

Problem 5

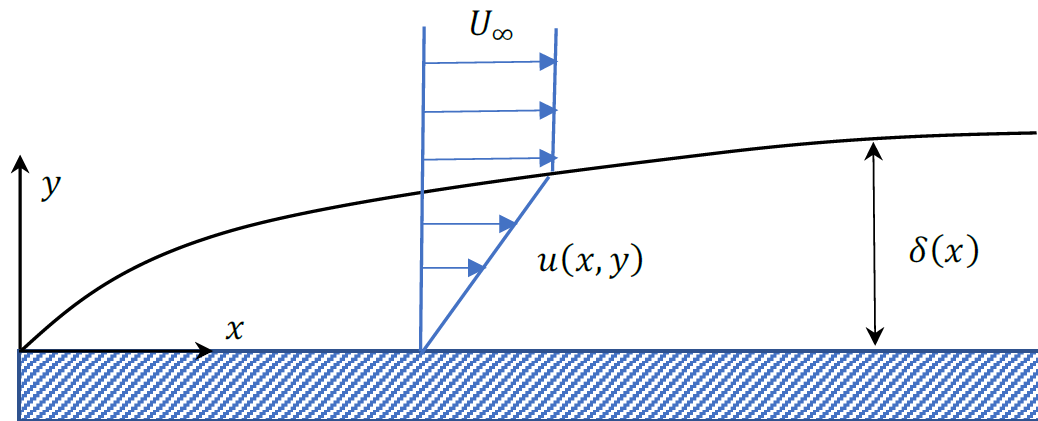
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A reasonable approximation for the two-dimensional incompressible boundary layer on the flat surface shown below is

$$u(x, y) = U_{\infty} \frac{y}{\delta}$$

where $u(x, y)$ is the horizontal velocity component, U_{∞} is the free stream velocity, δ is the boundary layer thickness given by $\delta = Cx^{1/2}$, and C is a

constant.



1. Explain why the velocity profile given above is reasonable.
2. What is the continuity equation for a two-dimensional incompressible flow?
3. Deduce that the vertical velocity component is given by

$$v(x, y) = \frac{1}{4} \frac{U_{\infty}}{C} y^2 x^{-3/2} + f(x)$$

where $f(x)$ is an arbitrary function of x .

4. Use the correct boundary condition to deduce $f(x)$.
5. At a distance of 1 m from the leading edge, the boundary layer thickness is 1 cm. Deduce the value of C .
6. Give an expression for the horizontal acceleration.
7. Calculate the horizontal acceleration at $x = 1$ m and $y = 0.5$ cm if the free stream velocity is 1 m/s.
8. Explain the difference between local and convective acceleration.
9. Calculate the (volumetric) flow rate through the boundary layer at $x = 1$ m.